

15-322 / 15-622 | FALL 2026

INTRODUCTION TO COMPUTER MUSIC

NOW
FULLY IN
PYTHON

— TURN CODE INTO SOUND



TUE / THU
11:00 AM–12:20 PM



CIC 1203



9 UNITS (15-322)
12 UNITS (15-622)

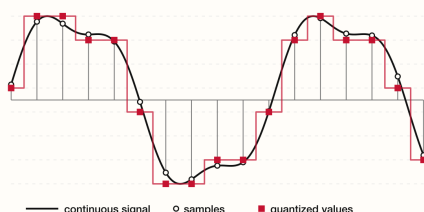
WHAT YOU'LL LEARN

DIGITAL AUDIO 01

SAMPLING · QUANTIZATION · ALIASING

$x(t) \rightarrow \text{sample} \rightarrow \text{3-bit quantize}$

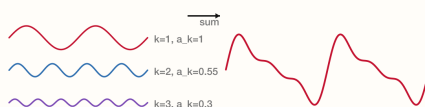
8 amplitude levels



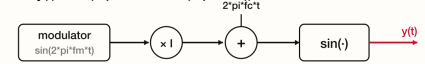
SYNTHESIS 02

ADDITIVE · FM · SAMPLE-BASED · PHYSICAL MODELING

ADDITIVE: $x(t) = \sum_k a_k \sin(2\pi k f_0 t)$



FM: $y(t) = \sin(2\pi f_c t) + I \sin(2\pi f_m t)$



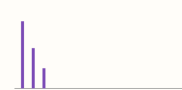
FOURIER & DSP 03

DFT / FFT · FILTERS · CONVOLUTION · STFT

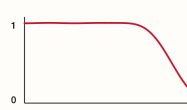
TIME DOMAIN



DFT MAGNITUDE



33-TAP WINDOWED-SINC LOW-PASS FIR



$y[n] = (x * h)[n]$

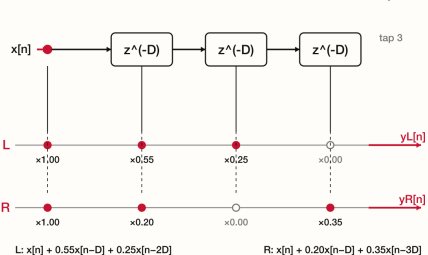
1 x Nyquist

EFFECTS & SPATIALIZATION 04

DELAY LINES · MODULATION · PHASE VOCODING

STEREO TAPPED DELAY LINE

D = one delay interval



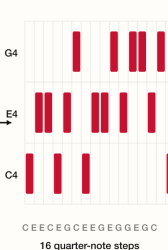
ALGORITHMIC COMPOSITION 05

MARKOV CHAINS · LPC · MIDI

MARKOV TRANSITION MATRIX

	C4	E4	G4
C4	0.10	0.60	0.30
E4	0.30	0.20	0.50
G4	0.60	0.30	0.10

SEQUENCE FROM SEED 322

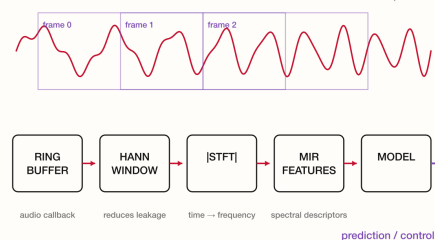


REAL-TIME AUDIO & MUSIC AI 06

STREAMING · MIR · MODELS

REAL-TIME FRAME PIPELINE

frame N=512 · hop H=256



9 WEEKLY ASSIGNMENTS
+ FINAL PROJECT



PREREQUISITE:
15-122 OR 15-112



MUSIC EXPERIENCE
NOT REQUIRED

INSTRUCTOR: CHRIS DONAHUE

Carnegie Mellon
University